Type MS · Name heart chambers and major vessels from a diagram

Answer 1 of 13 · Question 1 part a.i · 1 mark

2022 JAN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number					
1	(a)	(i)	B; (pulmonary artery) *A it is not the aorta* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(ii)	A; (aorta) *B it is not the pulmonary artery* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(iii)	in at top left and out through vessel; in at top right and out through vessel;		2
		(iv)	valves; prevent blood flowing other way/backflow;		2
	(b)	(i)	chamber width in cm P 1.0; Q 0.2; R 0.2; S 1.5;	1 correct = 1 mark 2 correct = 2 marks 3/4 correct = 3 marks	3
		(ii)	An answer that makes reference to three of the following: • The width corresponds to the amount of muscle; • S pumps blood furthest distance/whole body; • P pumps blood to lungs/short distance; • Q/R only receives blood/doesn't have to pump it very far/to ventricles;		3

Total : 12 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 1 of 13 · Question 1 part a.i · 1 mark

2022 JAN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 2 of 13 · Question 6 part a.i · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

6 (a)(i)	A aorta;		1
(ii)	D vena cava;		1
(iii)	• wider lumen; • thinner walls; • valves; 	Ignore arrows Accept valves drawn either way round	1 1 1
(iv)	Any three from: (Blood vessel X) • thicker walls/more muscle/elastic fibres in walls to maintain blood flow to maintain/withstand high (blood) pressure; • thinner lumen to maintain high blood pressure; OR (Blood vessel Y) - • less muscle/elastic tissue in walls so slower blood flow; • wider lumen so carries blood at lower pressure; • contains valves to prevent backflow;		Max 3

Type MS · Name heart chambers and major vessels from a diagram

Answer 2 of 13 · Question 6 part a.i · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

(b)	Any four from: • thin wall/wall one cell thick; • contains pores; • shorter diffusion pathway; • narrow lumen; • only one (blood) cell/few (blood) cells at a time can pass through;		Max 4

Total question 6 = 12 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 3 of 13 · Question 6 part a.ii · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

6 (a)(i)	A aorta;		1
(ii)	D vena cava;		1
(iii)	• wider lumen; • thinner walls; • valves; 	Ignore arrows Accept valves drawn either way round	1 1 1
(iv)	Any three from: (Blood vessel X) • thicker walls/more muscle/elastic fibres in walls to maintain blood flow to maintain/withstand high (blood) pressure; • thinner lumen to maintain high blood pressure; OR (Blood vessel Y) - • less muscle/elastic tissue in walls so slower blood flow; • wider lumen so carries blood at lower pressure; • contains valves to prevent backflow;		Max 3

Type MS · Name heart chambers and major vessels from a diagram

Answer 3 of 13 · Question 6 part a.ii · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

(b)	Any four from: • thin wall/wall one cell thick; • contains pores; • shorter diffusion pathway; • narrow lumen; • only one (blood) cell/few (blood) cells at a time can pass through;		Max 4

Total question 6 = 12 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 4 of 13 · Question 4 part a · 2 marks

2020 JAN · Paper 01 · find part a in this mark scheme extract

Question number				
4	(a)	• C (left) atrium(1) • D (left) ventricle (1)	Maximum of one mark if right atrium/ventricle given	2
	(b)	Any four from the following: • (blood vessels) A and F are arteries (1) • A is the aorta/F is the pulmonary artery (1) • arteries carry blood away from the heart (1) • blood vessels B and E are veins (1) • B is the vena cava/E is the pulmonary vein (1) • veins carry blood towards the heart (1)		4
	(c)	the heart pumps blood around the body/liver detoxifies the blood/involved in digestion / homeostasis/liver not muscular/liver has no chambers (1)		1
	(d)	(capillaries) exchange of materials(between the blood and body tissues/cells)/allows diffusion (1) named substance (1)		2

Total for Question 4 = 9 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 5 of 13 · Question 3 part a.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

Number				
3 (a)	(i)	A (1) (aorta) B it is not the pulmonary artery C it is not the pulmonary vein D it is not the vena cava		1
	(ii)	C (1) (right atrium) A it is not connected to the left atrium B it is not connected to the left ventricle D it is not connected to the right ventricle		1
(b)	(i)	circle drawn beyond by-pass vessel and around end of coronary vessels (1)		1
	(ii)	• no/less blood to heart muscles (1) • no/less oxygen/glucose transported (1) • no/less CO_2 removed (1)		3
	(iii)	• takes blood beyond blockage (1) • to supply heart muscles (1)		2

Total for question = 8 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 5 of 13 · Question 3 part a.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 6 of 13 · Question 3 part a.ii · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
3 (a)	(i)	A (1) (aorta) B it is not the pulmonary artery C it is not the pulmonary vein D it is not the vena cava		1
	(ii)	C (1) (right atrium) A it is not connected to the left atrium B it is not connected to the left ventricle D it is not connected to the right ventricle		1
(b)	(i)	circle drawn beyond by-pass vessel and around end of coronary vessels (1)		1
	(ii)	• no/less blood to heart muscles (1) • no/less oxygen/glucose transported (1) • no/less CO_2 removed (1)		3
	(iii)	• takes blood beyond blockage (1) • to supply heart muscles (1)		2

Total for question = 8 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 6 of 13 · Question 3 part a.ii · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 7 of 13 · Question 1 part a.ii · 1 mark

2022 JAN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number					
1	(a)	(i)	B; (pulmonary artery) *A it is not the aorta* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(ii)	A; (aorta) *B it is not the pulmonary artery* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(iii)	in at top left and out through vessel; in at top right and out through vessel;		2
		(iv)	valves; prevent blood flowing other way/backflow;		2
	(b)	(i)	chamber width in cm P 1.0; Q 0.2; R 0.2; S 1.5;	1 correct = 1 mark 2 correct = 2 marks 3/4 correct = 3 marks	3
		(ii)	An answer that makes reference to three of the following: • The width corresponds to the amount of muscle; • S pumps blood furthest distance/whole body; • P pumps blood to lungs/short distance; • Q/R only receives blood/doesn't have to pump it very far/to ventricles;		3

Total : 12 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 7 of 13 · Question 1 part a.ii · 1 mark

2022 JAN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 8 of 13 · Question 1 part a · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

1 (a)	pulmonary artery;		1
(b)	In the following order: • oxygenated; • lungs; • atrium; • ventricle; • body;		5
(c)	• blood vessel X has a thicker wall; • blood vessel X has a smaller lumen; • blood vessel X does not contain valves	Allow the reverse argument for blood vessel Y	3
			Total 9

Type MS · Name heart chambers and major vessels from a diagram

Answer 8 of 13 · Question 1 part a · 1 mark

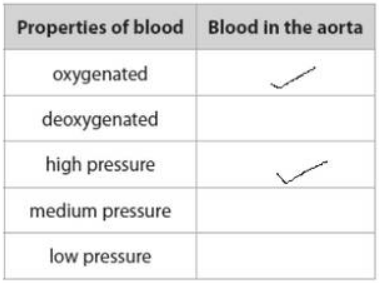
2022 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 9 of 13 · Question 2 part a.i · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
2	(a)	(i)	left ventricle;		1
		(ii)	Two from: • valve; • prevents the backflow of blood; • so the blood flows the correct way around the heart/in one direction;	Ignore names of valves	2
		(iii)	 Properties of blood / Blood in the aorta oxygenated ✓ deoxygenated high pressure ✓ medium pressure low pressure	One mark removed for every additional incorrect tick.	2
		(iv)	the lungs/pulmonary vein;		1
	(b)		Two from: • it could be diseased/damaged/fatty deposits/blocked arteries; • the tissue/blood type doesn't match; • it has been removed from person for too long; • wrong size;	accept named diseases allow risk of rejection	2
	(c)	(i)	C platelets;		1
		(ii)	• fibrinogen is not converted to fibrin/fibrin not produced; • so factor XIII has no substrate/fibrin cannot form the clot/mesh not formed;		2

Total 11 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 9 of 13 · Question 2 part a.i · 1 mark

2023 JAN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 10 of 13 · Question 7 part a · 1 mark

2023 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

7 (a)	pulmonary artery;		1
(b)	• Y is thicker; • blood pumped into aorta; • takes/distributes blood to organs/tissues/whole body; • requires high pressure • thicker wall means more muscle for stronger contraction;	ORA for each point	5
(c)	any three of • heart muscle deprived of oxygen; • insufficient energy for muscle contraction/less aerobic respiration/(more) anaerobic respiration; • heart stops pumping blood/pumps less blood; • heart attack;		3
(d)	• transfers CO_2 (from muscle cells) to blood capillary; • transfers oxygen from blood/capillary (to muscle cells); • transfers glucose from blood/capillary (to muscle cells);		3

Total 12 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 10 of 13 · Question 7 part a · 1 mark

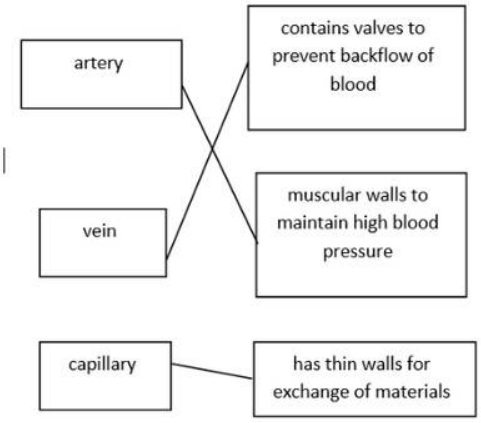
2023 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

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Type MS · Name heart chambers and major vessels from a diagram

Answer 11 of 13 · Question 2 part c.i · 1 mark

2024 JUN · Paper 01 (regional R) · find part c.i in this mark scheme extract

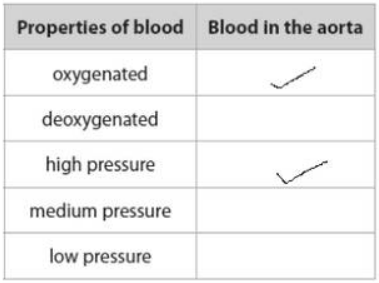
2 (a) (i)	X = plasma; Y = red blood cell; Z = white blood cell/phagocyte;	Accept named phagocyte	(3)
(a) (ii)	• transports oxygen; • to body/ cells/tissues/organs;		(2)
(a) (iii)	the patient has an infection/leukaemia/allergy/ blood cancer, inflammation;	Accept pathogens in body	(1)
(b)		Reject more than one line from each blood vessel 3 correct=2marks 1 or 2 correct= 1mark	(2)
(c) (i)	left ventricle;		(1)
(c) (ii)	• into right ventricle; • through pulmonary artery; • to the lungs; • (from the lungs) into the pulmonary vein;		(4)

Total for Question 2 = 13 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 12 of 13 · Question 2 part a.iv · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part a.iv in this mark scheme extract

Number					
2	(a)	(i)	left ventricle;		1
		(ii)	Two from: • valve; • prevents the backflow of blood; • so the blood flows the correct way around the heart/in one direction;	Ignore names of valves	2
		(iii)	 Properties of blood / Blood in the aorta oxygenated ✓ deoxygenated high pressure ✓ medium pressure low pressure	One mark removed for every additional incorrect tick.	2
		(iv)	the lungs/pulmonary vein;		1
	(b)		Two from: • it could be diseased/damaged/fatty deposits/blocked arteries; • the tissue/blood type doesn't match; • it has been removed from person for too long; • wrong size;	accept named diseases allow risk of rejection	2
	(c)	(i)	C platelets;		1
		(ii)	• fibrinogen is not converted to fibrin/fibrin not produced; • so factor XIII has no substrate/fibrin cannot form the clot/mesh not formed;		2

Total 11 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 12 of 13 · Question 2 part a.iv · 1 mark

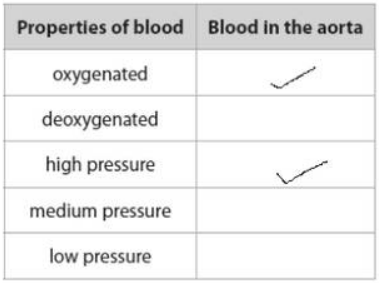
2023 JAN · Paper 02 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 13 of 13 · Question 2 part a.iii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number					
2	(a)	(i)	left ventricle;		1
		(ii)	Two from: • valve; • prevents the backflow of blood; • so the blood flows the correct way around the heart/in one direction;	Ignore names of valves	2
		(iii)	 Properties of blood / Blood in the aorta oxygenated ✓ deoxygenated high pressure ✓ medium pressure low pressure	One mark removed for every additional incorrect tick.	2
		(iv)	the lungs/pulmonary vein;		1
	(b)		Two from: • it could be diseased/damaged/fatty deposits/blocked arteries; • the tissue/blood type doesn't match; • it has been removed from person for too long; • wrong size;	accept named diseases allow risk of rejection	2
	(c)	(i)	C platelets;		1
		(ii)	• fibrinogen is not converted to fibrin/fibrin not produced; • so factor XIII has no substrate/fibrin cannot form the clot/mesh not formed;		2

Total 11 marks

Type MS · Name heart chambers and major vessels from a diagram

Answer 13 of 13 · Question 2 part a.iii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks